

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network
Security COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks: 50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

(192425207)
I D. Arund paul (192425207) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- Deduce the most likely original message.
- Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- Demonstrate how the encryption process transforms the message step-by-step.
- Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- Analyze how investigators could detect the presence of hidden data within the image.
- Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of messages in this scenario.

1. Event Communication Security

A student organization wants to share confidential information about an upcoming event with its core members. To protect the message from unauthorized people, they use a simple classical encryption technique called the Caesar cipher.

original message: MEET AT NOON

Encryption:— This can be done using a Caesar cipher, where each letter is shifted 4 positions forward in the alphabet.

M → Q

E → I

E → I

T → X

A → E

T → X

N → R

O → S

O → S

N → R.

Therefore:—

original: MEET AT NOON

Encrypted: QIX EX RSSR.

Decryption:— An authorized member who knows the shift value 4 shifts every encrypted letter 4 positions backward.

Q 11x Ex NSSR → MEET AT NOON.

Conclusion:- Caesar cipher provides simple message encryption using a fixed shift value.

24. Scenario : Suspicious Digital Asset Investigation.

Answer:- Investigators can check the logo image for steganography, which is the process of hiding information inside another file.

Steganography is a method of hiding information inside another file, such as an image, audio file, video, or document. The main purpose is to make the existence of the hidden information.

Another important method is pixel analysis. Digital images are made up of pixels containing colour values. Secret information can be hidden by making very small changes to these values.

1. Encryption before hiding:-

The confidential message should first be encrypted using a strong encryption algorithm using a strong encryption algorithm. The encrypted message can investigators extract the hidden data.

2. Randomized embedding:-

Instead of hiding information in a simple or predictable sequence of pixels, the data can be distributed across randomly selected pixels using a secret key. This makes statistical detection and extraction more difficult.

Therefore, investigators can combine metadata analysis, pixel analysis, file comparison, and forensic tools to detect hidden information.

Conclusion:-

Steganography can make confidential communication appear like an ordinary image. However, careful forensic analysis can reveal unusual patterns or hidden data. Encryption and randomized embedding can provide additional protection for hidden communications.